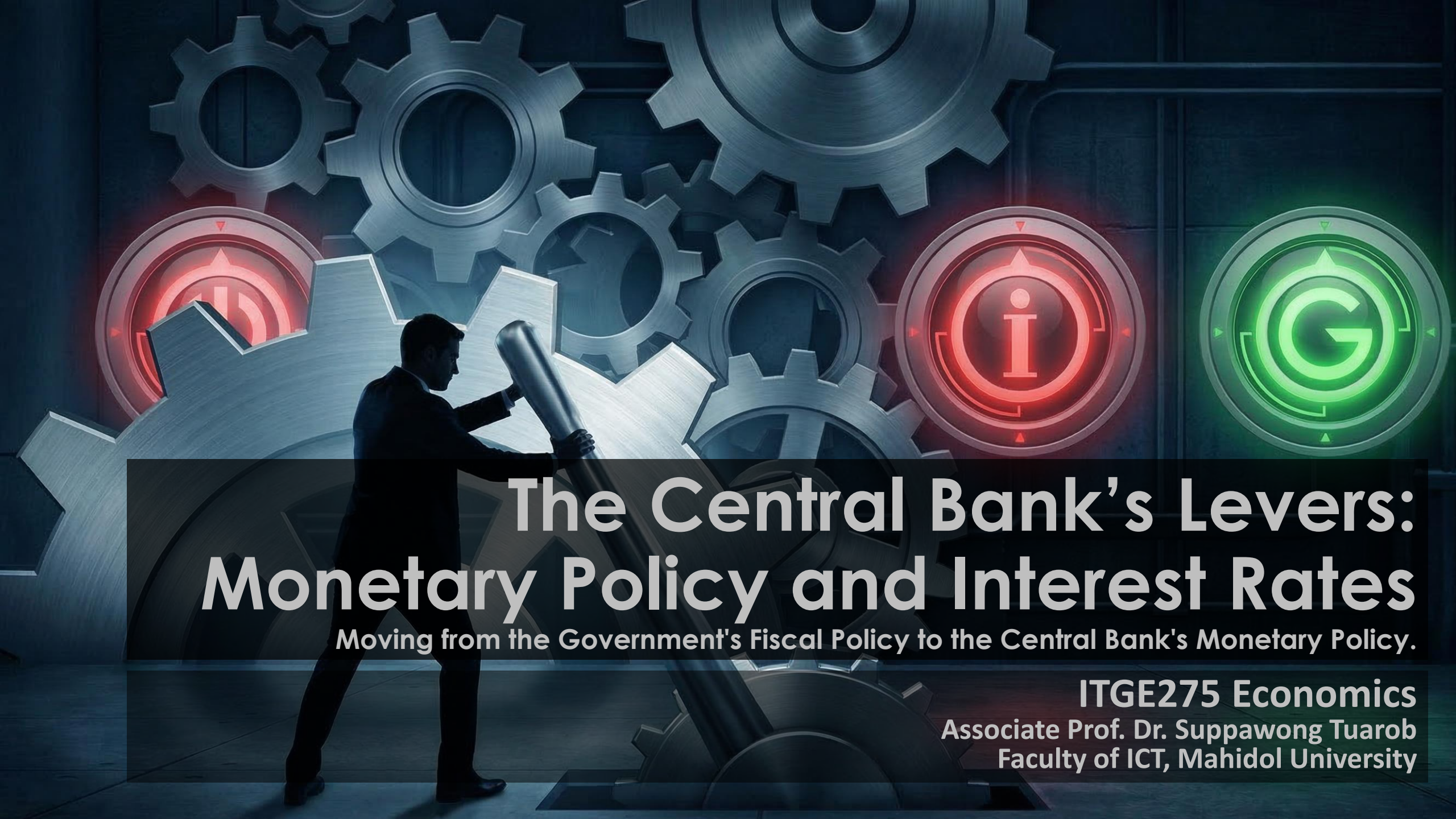


# The Central Bank's Levers: Monetary Policy and Interest Rates

Moving from the Government's Fiscal Policy to the Central Bank's Monetary Policy.

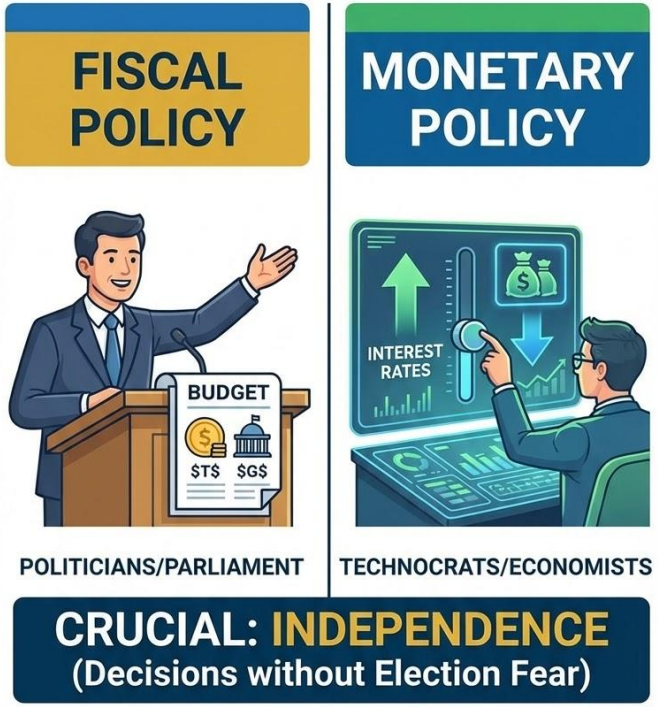

ITGE275 Economics

Associate Prof. Dr. Suppawong Tuarob

Faculty of ICT, Mahidol University

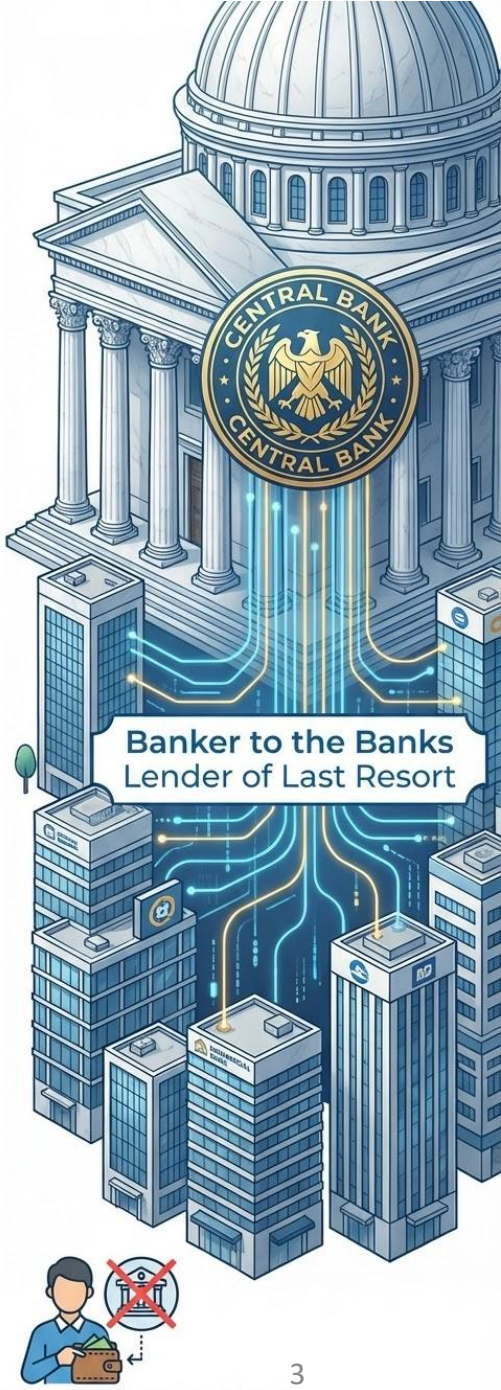
# The Two Managers of the Economy

- In Session 9, we learned how the Government steers the economy through Fiscal Policy (Taxes and Spending).
- Today, we meet the other independent manager: **The Central Bank**.
- **Fiscal Policy:** Controlled by Politicians/Parliament. Tools: Government Budgets (T, G).
- **Monetary Policy:** Controlled by Technocrats/Economists. Tools: Money Supply and Interest Rates.
- Independence is crucial: Central Banks must often make unpopular decisions (like raising rates) without fear of losing an election.



# What is a Central Bank?

- **Definition:** A national or regional institution responsible for managing a country's currency, money supply, and interest rates.
- It is the "Banker to the Banks" and the "Lender of Last Resort."
- **Examples:**
  - **Thailand:** The Bank of Thailand (BoT).
  - **United States:** The Federal Reserve (The "Fed").
  - **Europe:** The European Central Bank (ECB).
- Unlike commercial banks, they do not deal directly with regular consumers. You cannot open a savings account at the Bank of Thailand.



# The Dual Mandate

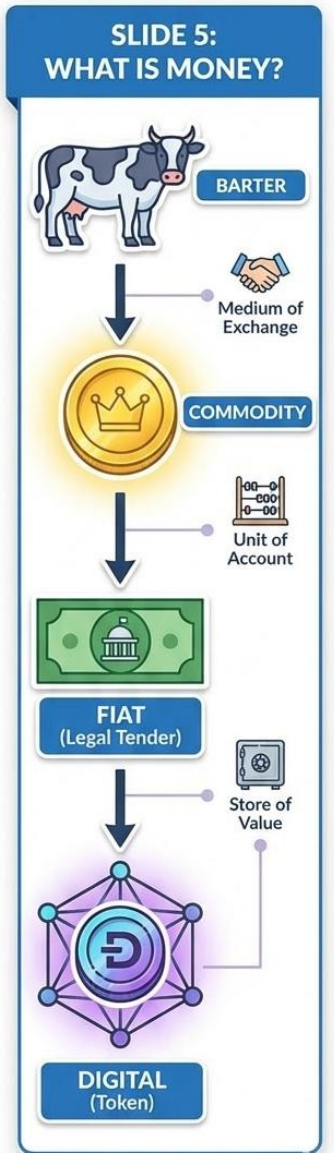
- Central Banks generally have two primary (and often conflicting) objectives:
  - **Price Stability:** Keeping inflation low, stable, and predictable (usually targeting around 2% per year).
  - **Maximum Sustainable Employment:** Keeping the economy growing fast enough to create jobs without overheating.
- **The Balancing Act:** Stimulating growth often increases inflation. Fighting inflation often slows down growth and costs jobs. This is the ultimate macroeconomic trade-off.



Part 2: Money and The Banking System

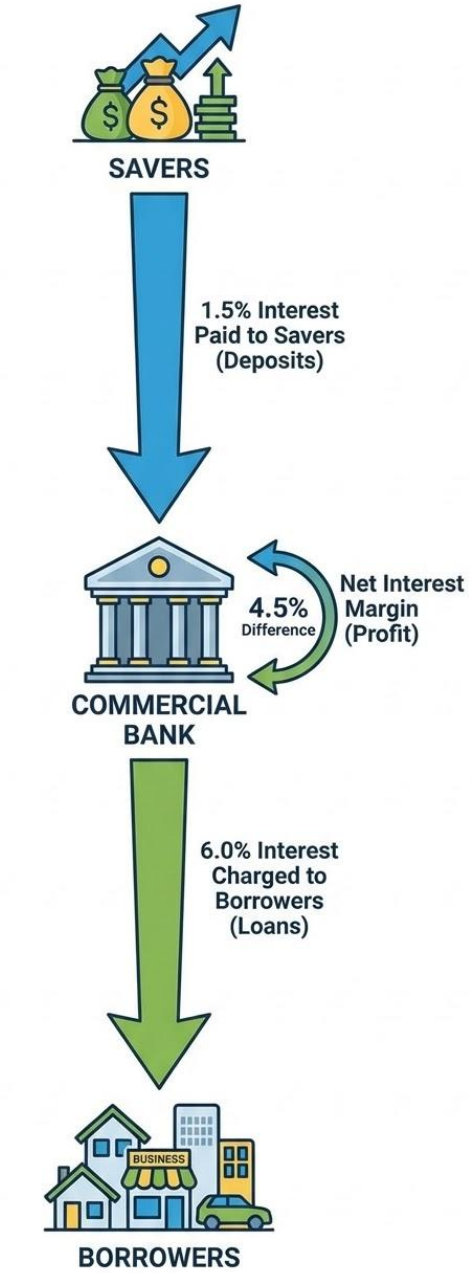
# What is Money?

- Before understanding how to manage money, we must define it.
- **Fiat Currency:** Everyday money (like paper bills) isn't backed by gold and is essentially worthless on its own. It only holds value for two reasons: the government says it's official money, and we all agree to trust it.
- **Functions of Money:**
  - *Medium of Exchange:* Used to buy and sell.
  - *Unit of Account:* A standard numerical metric of value.
  - *Store of Value:* Retains purchasing power over time (if inflation is low).



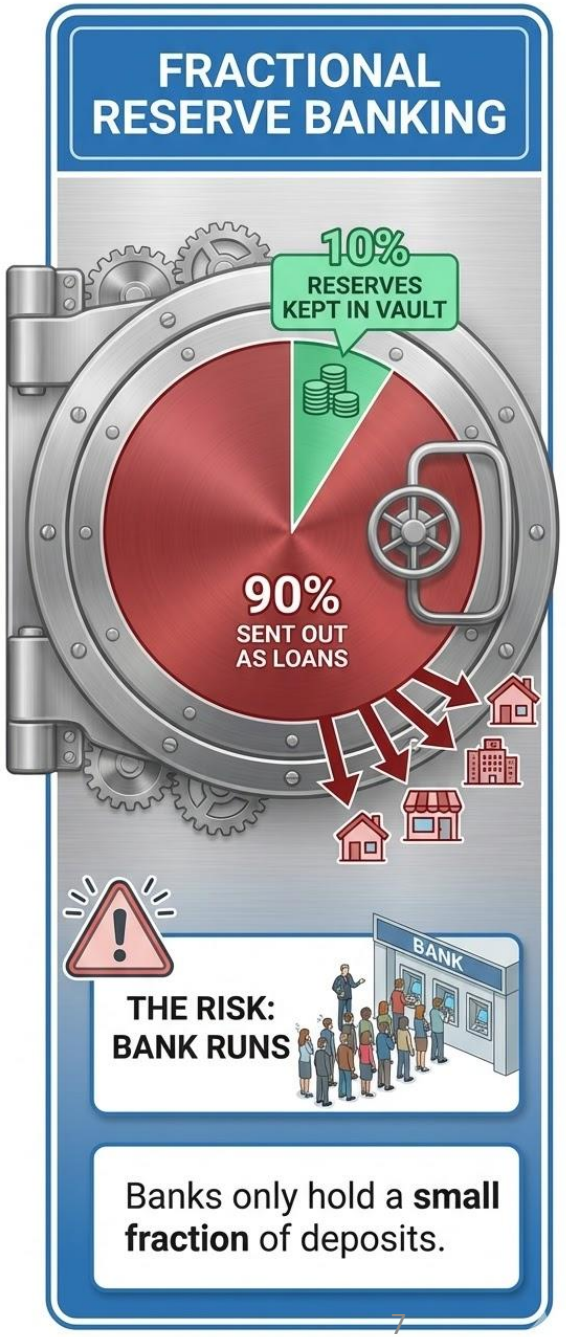
# The Commercial Banking System

- **Role:** Commercial banks (like SCB, KBank, JP Morgan) act as intermediaries between Savers and Borrowers.
- **The Business Model:**
  - They take in deposits from people with excess funds (paying them a low interest rate).
  - They lend those funds to people/businesses who need capital (charging them a higher interest rate).
  - The difference between these rates is the "Net Interest Margin" (their profit).



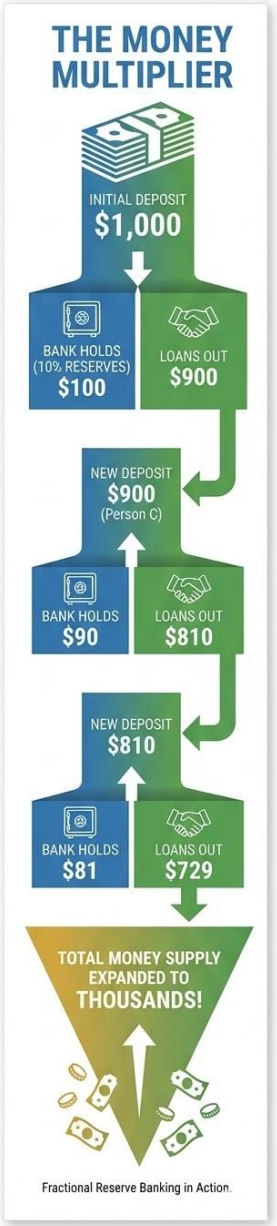
# Fractional Reserve Banking

- Banks do not keep 100% of deposits in their vaults. If they did, they couldn't make loans or pay interest.
- **The Rule:** They are only required to hold a small "fraction" of their deposits as reserves (e.g., 10%) to handle daily withdrawals.
- **The Risk:** Bank Runs. If everyone tries to withdraw their money at the exact same time, the bank will not have enough cash on hand.



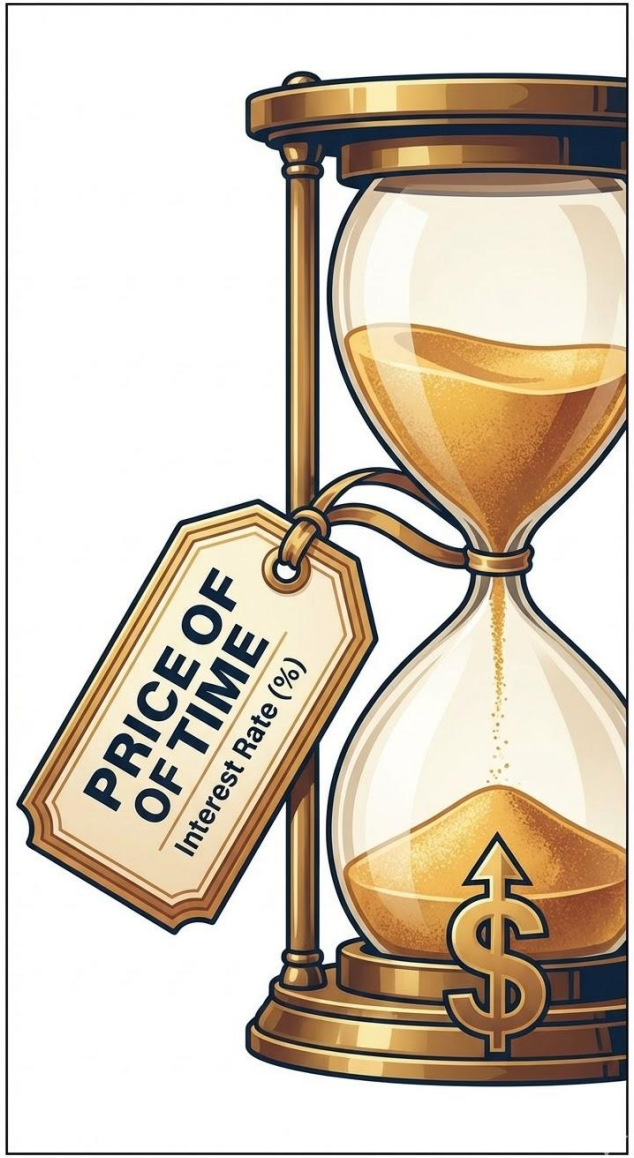
# The Money Multiplier

- Because of fractional reserve banking, commercial banks literally *create* money.
- **The Process:**
  - Person A deposits $1,000.
  - Bank holds $100, loans out $900 to Person B.
  - Person B buys a laptop from Person C for $900.
  - Person C deposits $900 in their bank.
  - Bank holds $90, loans out $810.
- **Result:** The initial $1,000 deposit expanded the total money supply in the economy to thousands of dollars.



# The Cost of Money: Interest

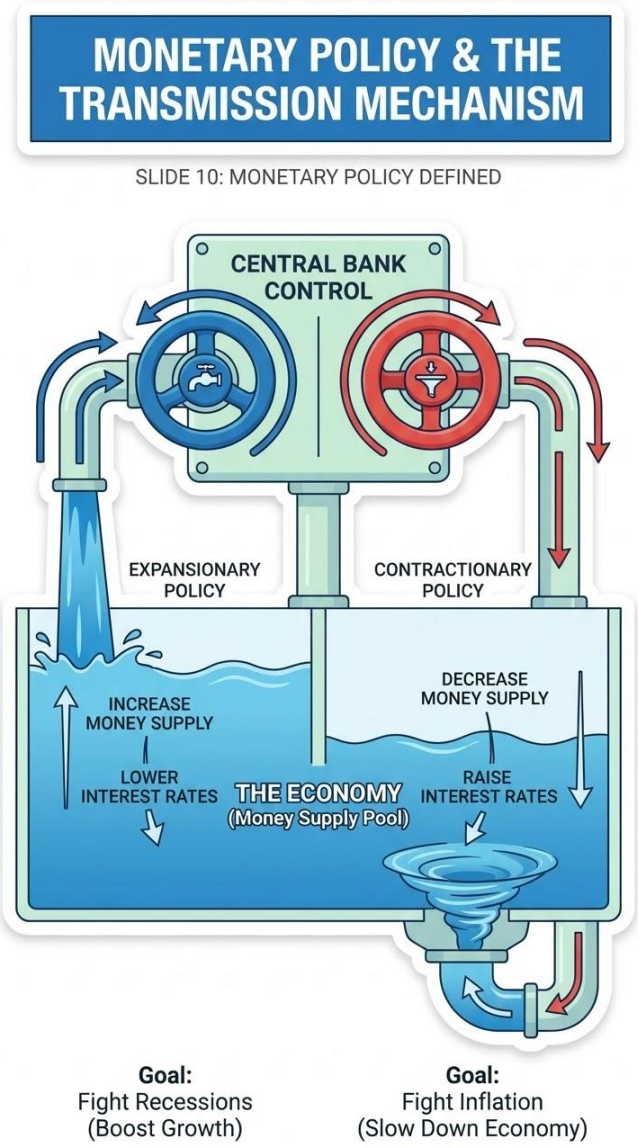
- Because money can be invested to generate returns, and because lending involves the risk of default (ความเสี่ยงในการผิดนัดชำระ), borrowing money comes with a price tag.
- **Interest Rate:** The cost of borrowing money, or the reward for saving it, expressed as a percentage.
- If the Central Bank wants to slow down the creation of money (the multiplier effect), they must make borrowing more expensive.



Part 3: Monetary Policy & The Transmission Mechanism

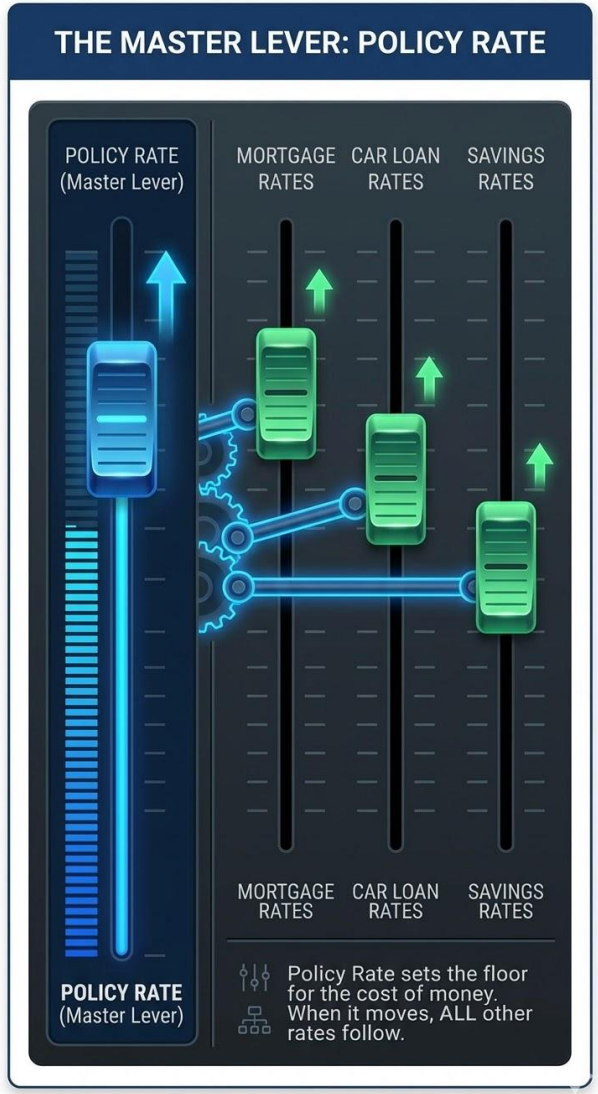
# Monetary Policy Defined

- **Definition:** The actions of a central bank that determine the size and rate of growth of the money supply, which in turn affects interest rates.
- **Expansionary Policy:** Used to fight recessions. Goal is to increase money supply and lower interest rates.
- **Contractionary Policy:** Used to fight inflation. Goal is to decrease money supply and raise interest rates.



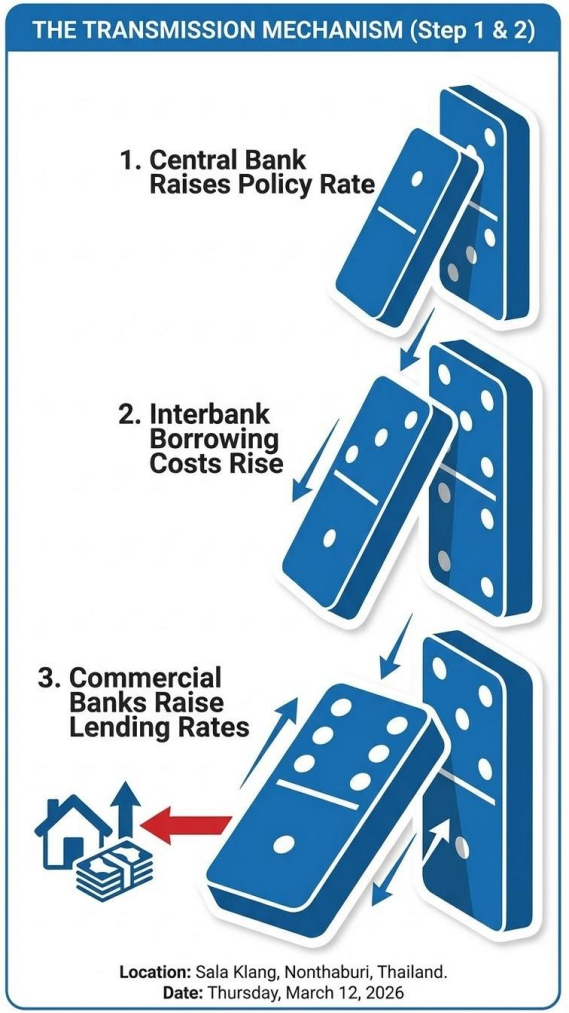
# The Master Lever: The Policy Rate

- The Central Bank does not dictate the mortgage rate you pay. It sets the **Policy Rate** (e.g., the Federal Funds Rate, or the BoT 1-day Repurchase Rate).
- **What is it?** It is the interest rate at which commercial banks lend reserve balances to *each other* overnight.
- If a bank is short on cash at the end of the day, it borrows from another bank. The Policy Rate sets the floor for the cost of money in the entire economy.



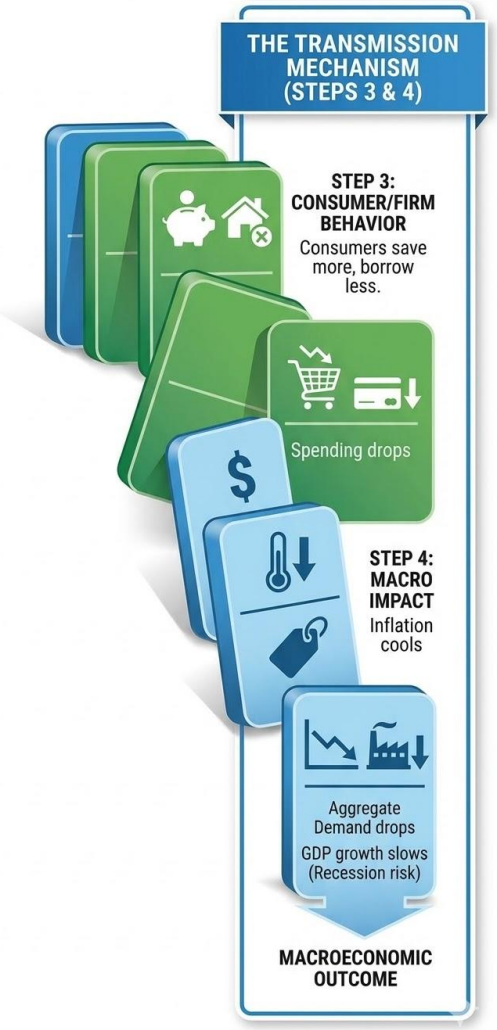
# The Transmission Mechanism (Step 1 & 2)

- How does changing an overnight bank rate affect you?
- **Step 1 (Interbank):** Central Bank raises the Policy Rate. It is now more expensive for commercial banks to borrow money to meet their reserve requirements.
- **Step 2 (Commercial Rates):** To maintain their profit margins, commercial banks immediately pass this cost onto consumers. They raise their Prime Lending Rates (for businesses) and Mortgage Rates (for consumers).



# The Transmission Mechanism (Step 3 & 4)

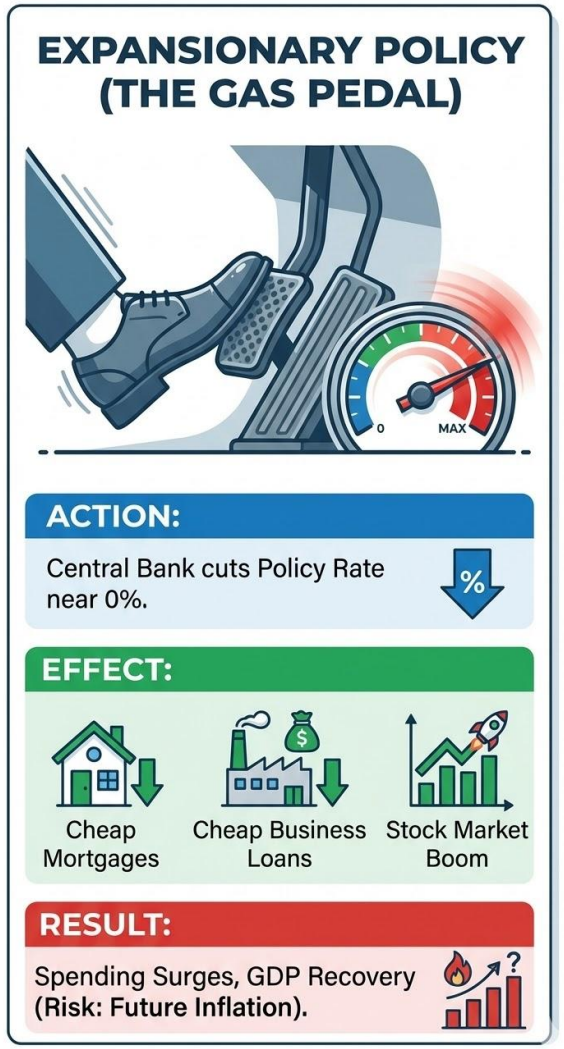
- **Step 3 (Consumer/Firm Behavior):** * *Borrowing is expensive:* You delay buying a house. A company delays building a new factory.
  - *Saving is lucrative:* Higher deposit rates encourage people to leave money in the bank rather than spend it.
- **Step 4 (Macro Impact):** Aggregate Demand drops. Because fewer people are buying goods, businesses stop raising prices (Inflation drops). If demand drops too much, GDP growth slows (Recession risk).



# Expansionary Policy (The Gas Pedal)

- **Scenario:** The economy crashes (e.g., COVID-19 in 2020).
- **Action:** Central Bank *cuts* the Policy Rate near 0%.
- **Effect:** * Mortgages become incredibly cheap.
  - Businesses can borrow money for almost nothing.
  - The stock market booms as investors look for higher returns outside of low-yield savings accounts.
- **Result:** Spending surges, stimulating GDP recovery (but risking future inflation).

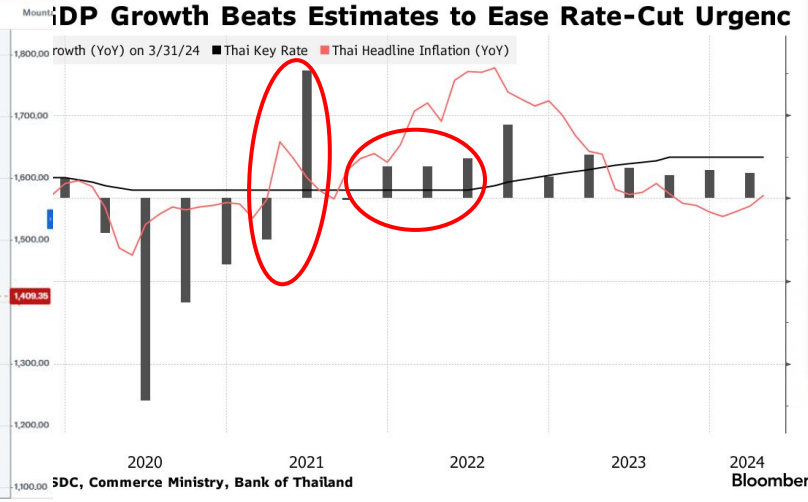


# Contractionary Policy (The Brakes)

- **Scenario:** The economy overheats (e.g., Post-COVID 2022).
- **Action:** Central Bank aggressively *raises* the Policy Rate.
- **Effect:**
  - Credit card debt and mortgages become highly expensive.
  - Companies lay off workers to cut costs due to expensive capital.
  - Money leaves the stock market and goes into high-yield safe bank accounts.
- **Result:** Spending halts, inflation cools down, but GDP growth stalls.






## Part 4: Computational Approach: Interest Rates

# Simple vs. Compound Interest

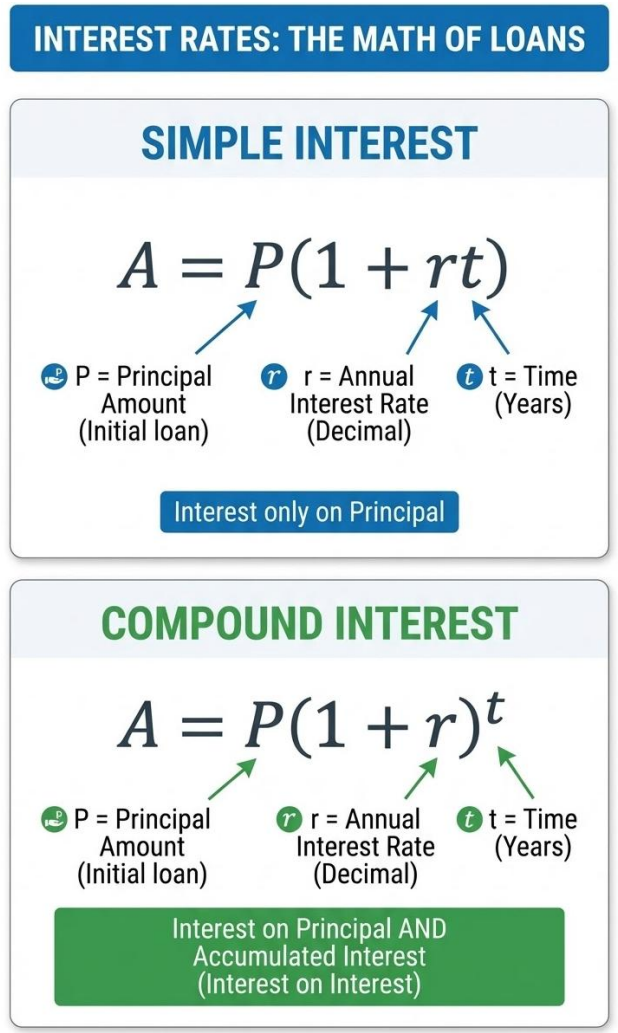
- To understand why rate hikes hurt so much, we must look at the math of loans.
- **Simple Interest:** Interest is calculated only on the principal amount.

$$A = P(1 + rt)$$

- **Compound Interest:** Interest is calculated on the principal *and* the accumulated interest of previous periods. (Interest on interest).

$$A = P(1 + r)^t$$

- All major consumer loans (mortgages) and investments rely on compound interest. Over 30 years, the difference is astronomical.



# Visualizing Interest in Python

- In our Jupyter Notebook, we build a plot_interest_types function to visualize this impact.
- **Inputs:** * Principal Loan Amount
  - Interest Rate (%)
  - Loan Term (Years)
- We use numpy arrays to calculate the growth year-by-year and matplotlib to plot the curves.

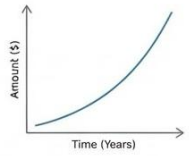
```
import numpy as np
import matplotlib.pyplot
as plt

# Inputs
principal = 10000
rate = 0.05  # 5%
years = 10

# Numpy array for time
time_array = np.arange(0,
years + 1)

# Compound Interest
Calculation
compound_interest =
principal * (1 + rate)**time_array

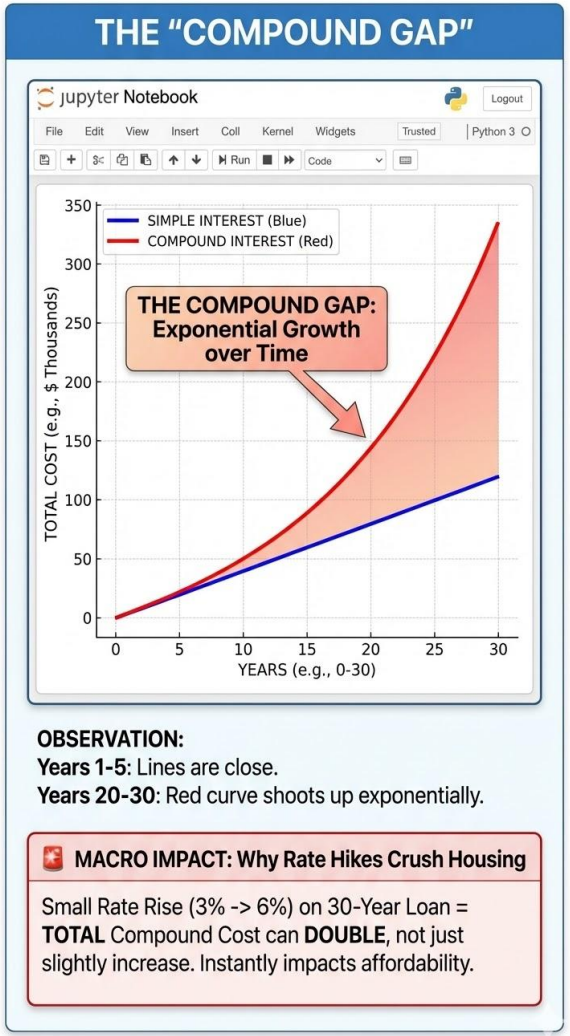
# Plotting
plt.plot(time_array,
compound_interest)
plt.show()
```




Visualizing Interest Growth in Python

# The "Compound Gap"

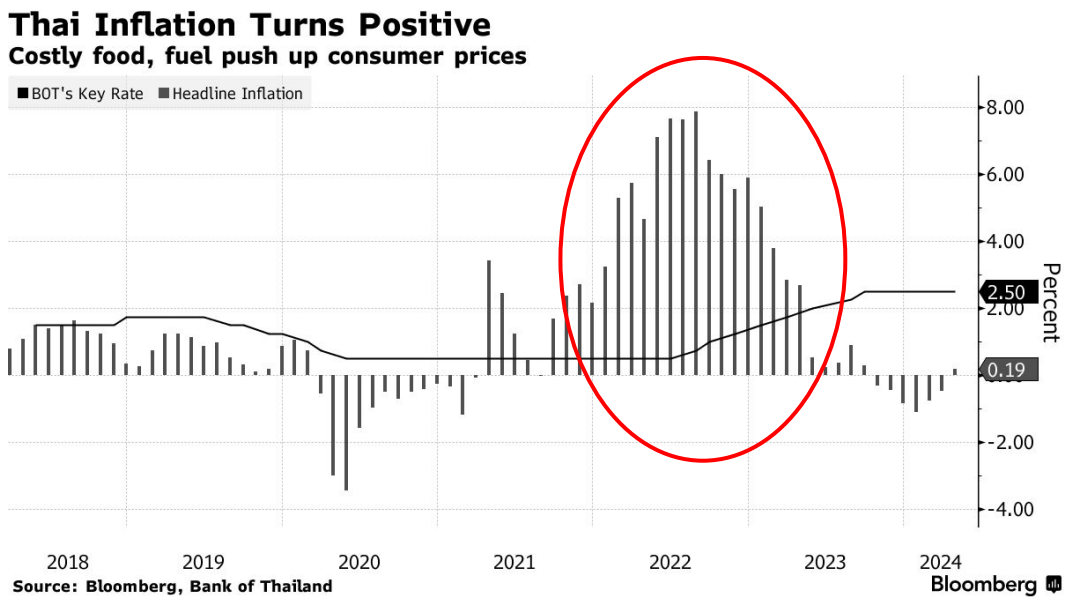
- The notebook output reveals the "Compound Gap."
- **Observation:** In the first 5 years, the blue line (Simple) and red line (Compound) are close. By year 20 or 30, the red curve shoots upward exponentially.
- **Macro Impact:** When central banks raise rates from 3% to 6%, a 30-year mortgage doesn't just get slightly more expensive; the total compound cost over the life of the loan can easily double, instantly crushing the housing market.

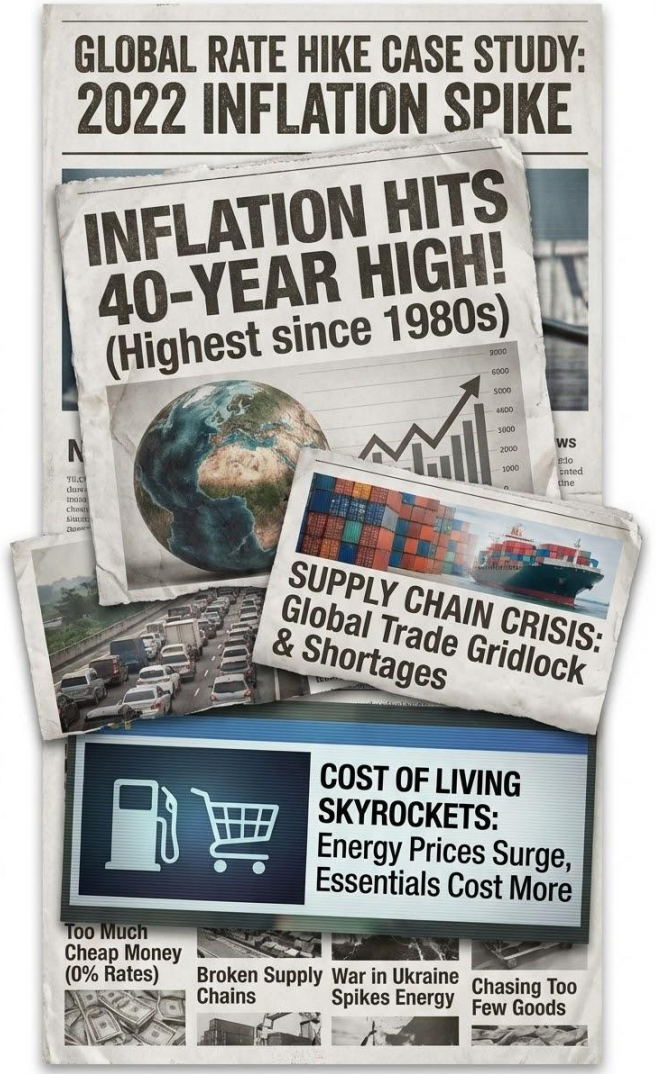


Part 5: Case Study: 2022-2024 Global Rate Hike

# Context: The 2022 Inflation Spike

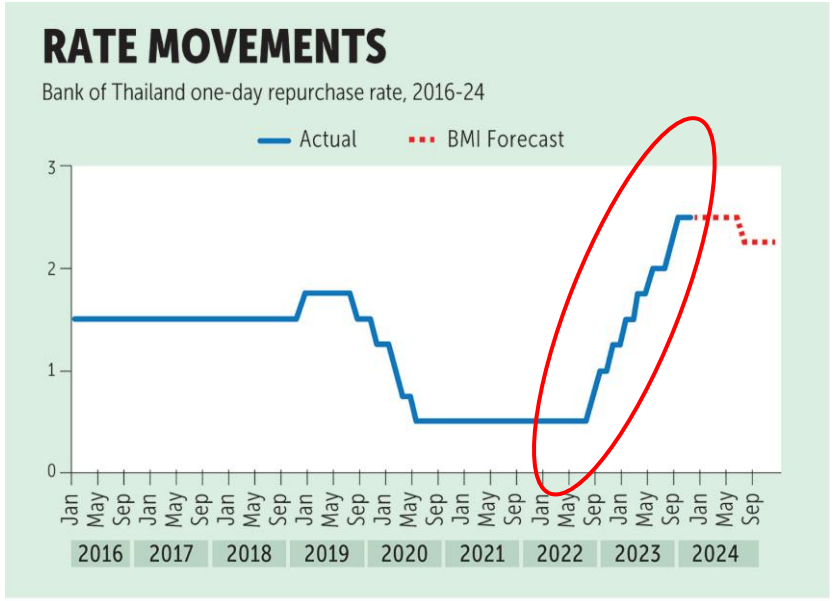
- Coming out of the pandemic, central banks kept rates at 0% to help the economy.
- However, supply chains were broken, and the war in Ukraine caused energy prices to spike.
- **The Result:** The highest inflation seen in 40 years. Too much cheap money (0% rates) chasing too few goods (broken supply chains).

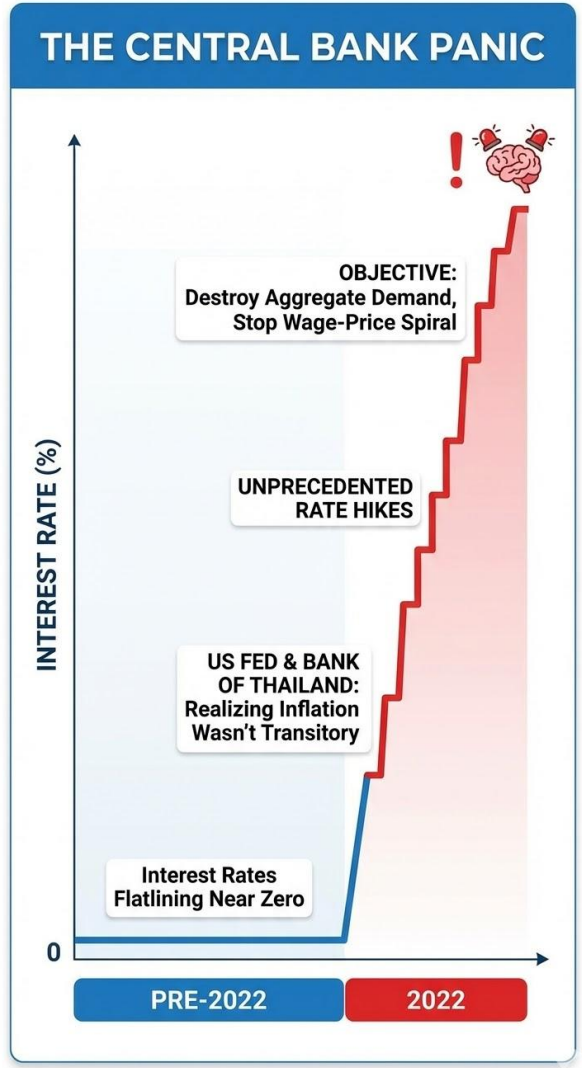


# The Central Bank *Panic*

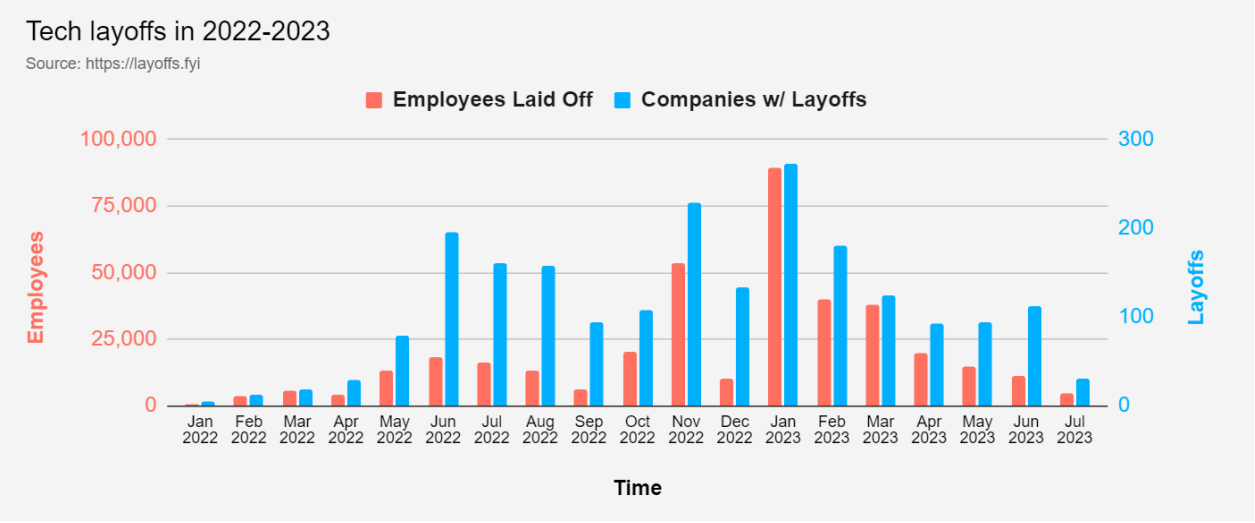
- Realizing inflation wasn't "transitory," central banks panicked.
- The US Federal Reserve and the Bank of Thailand began hiking rates at an unprecedented speed.
- **The Objective:** Destroy aggregate demand fast enough to stop a wage-price spiral (where inflation becomes a permanent psychological expectation).



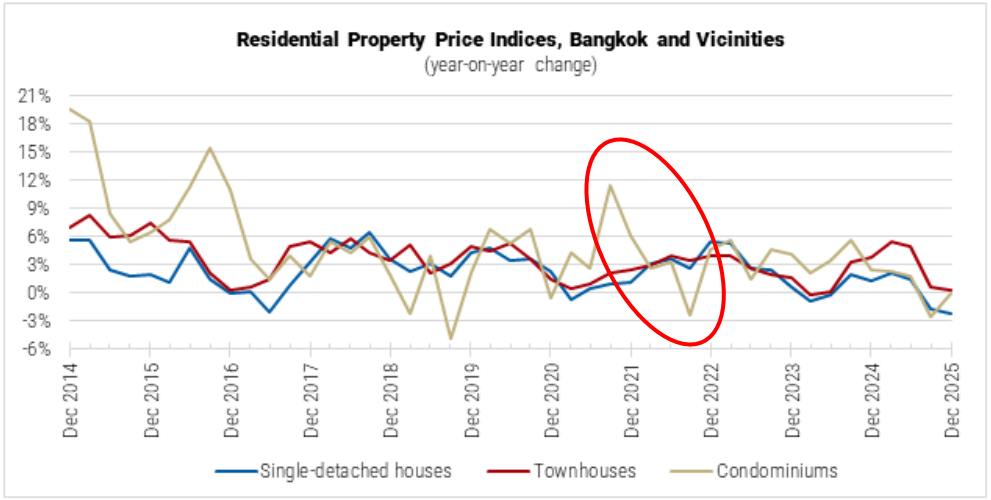



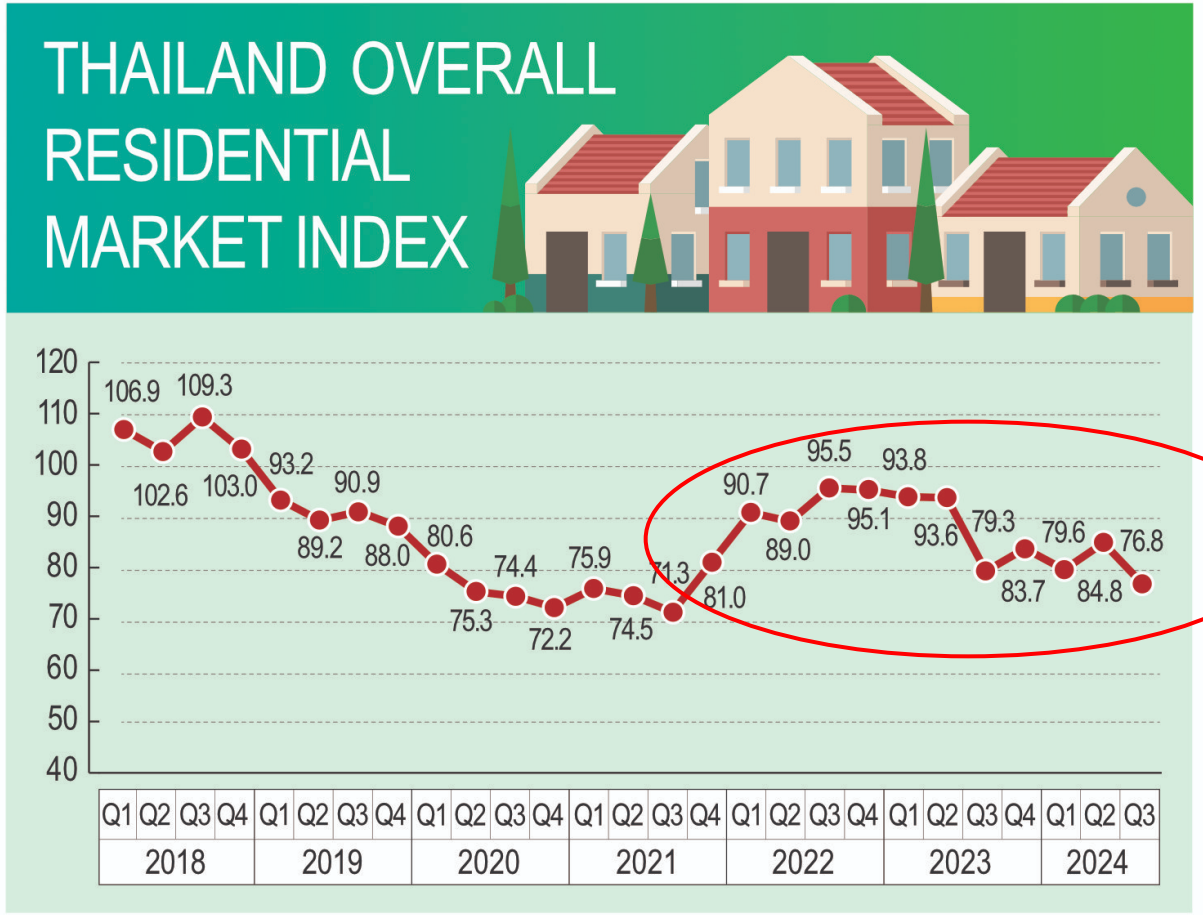
# The Collateral Damage (Tech & Real Estate)

- **Tech Stocks:** Tech companies rely on borrowing cheap money to fund years of un-profitable growth. When money became expensive, tech valuations crashed, leading to massive layoffs in 2022/2023.
- **Real Estate:** Mortgage rates jumped from 3% to 7%. The monthly payment on a standard home doubled. Home buying froze.
- This is the Transmission Mechanism working exactly as intended: destroying demand to cool prices.





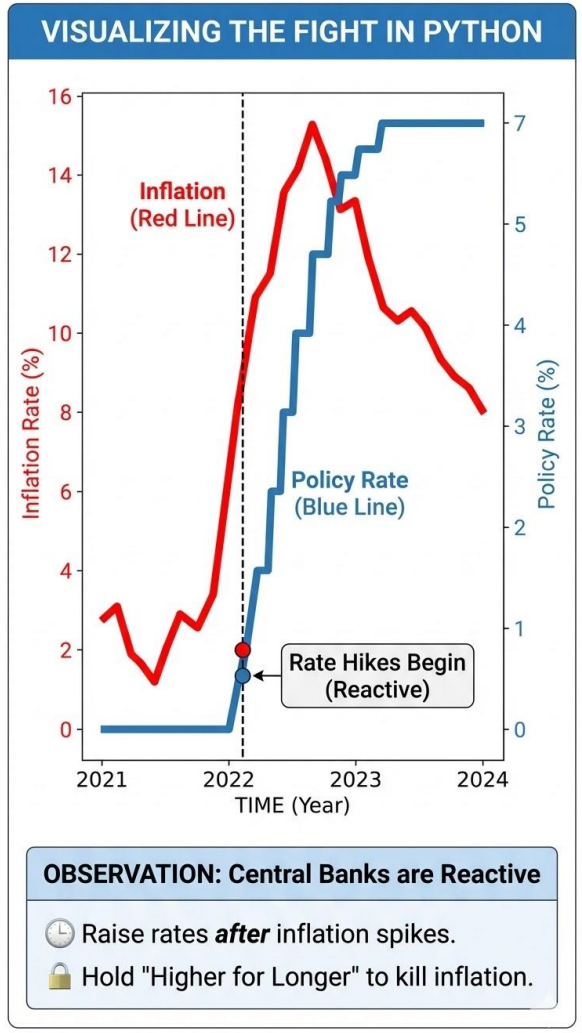




Source: Real Estate Information Center

BANGKOK POST GRAPHICS

# Visualizing the Fight in Python

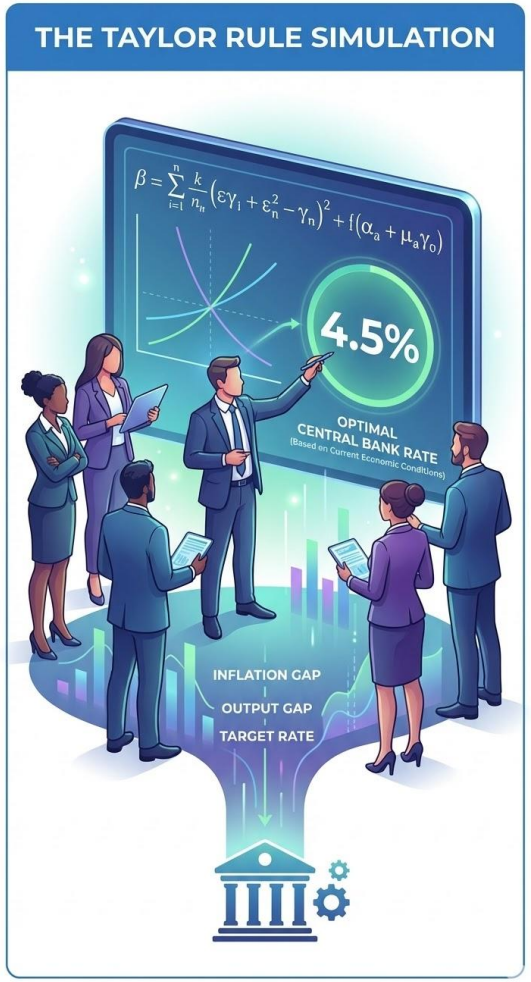
- In the notebook, we generate a dual-axis chart to map this specific timeline.
- **Red Line (Inflation):** Spikes in 2022, peaks, and slowly starts falling.
- **Blue Line (Policy Rate):** Flat at 0.5%, then sharply stair-steps upward, chasing the red line.
- **Observation:** Central Banks are reactive. They raise rates *after* inflation spikes, and hold them "Higher for Longer" until inflation is dead.



Part 6: The Taylor Rule Simulation

# How Do Central Banks Decide the Rate?

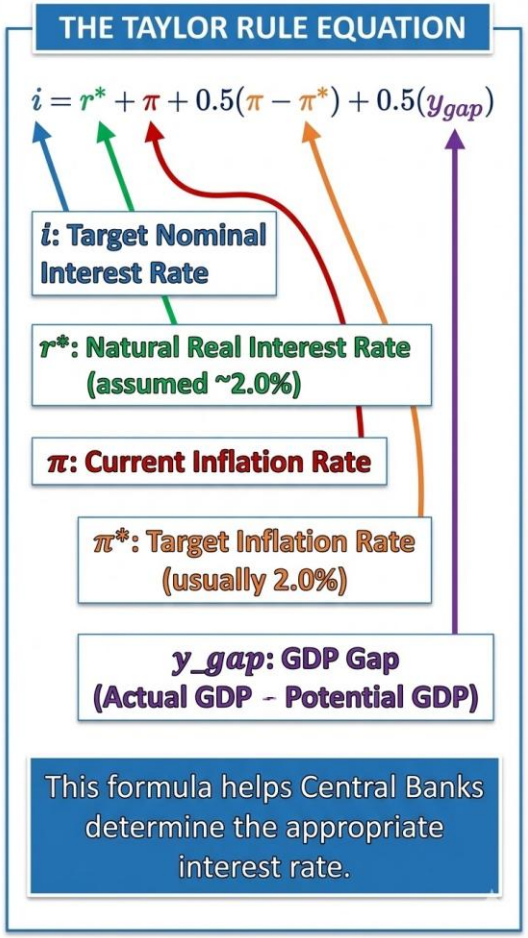
- The Central Bank committee doesn't just guess a number. They rely on complex macroeconomic models and heuristics.
- The most famous heuristic is the **Taylor Rule**, proposed by economist John Taylor in 1993.
- It is a mathematical formula that suggests what the "optimal" central bank interest rate should be based on current economic conditions.



# The Taylor Rule Equation

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5(y_{gap})$$

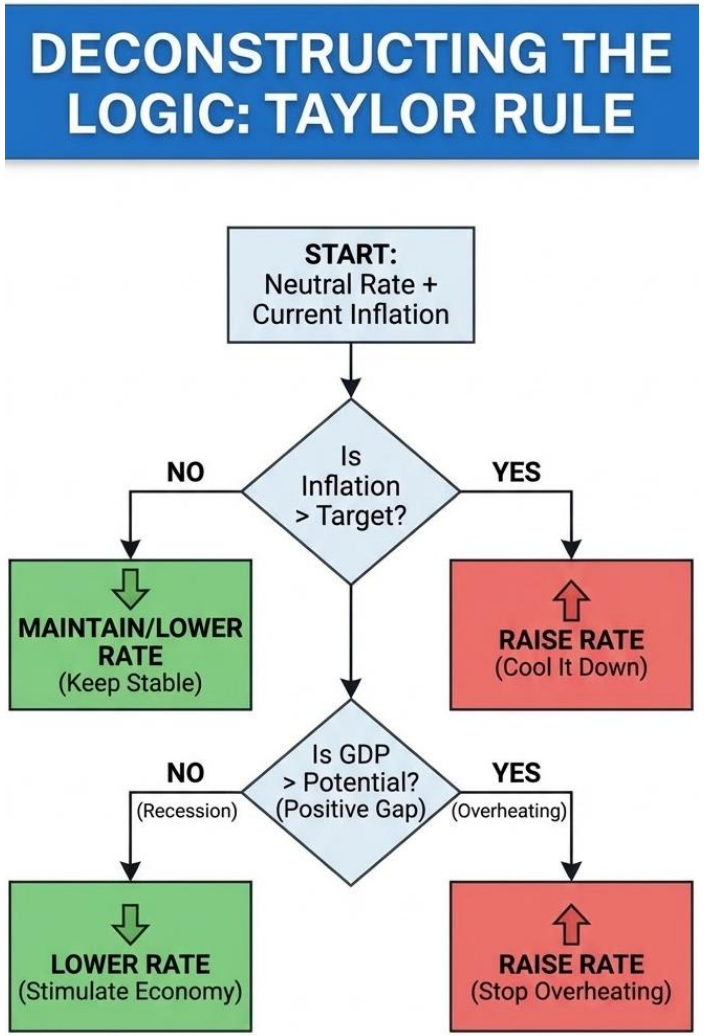
- **i**: Target Nominal Interest Rate.
- **r***: Natural Real Interest Rate (assumed ~2.0%).
- π: Current Inflation Rate.
- π* : Target Inflation Rate (usually 2.0%).
- $\mathbf{y_{gap}}$: GDP Gap (Actual GDP - Potential GDP).



# Deconstructing the Logic

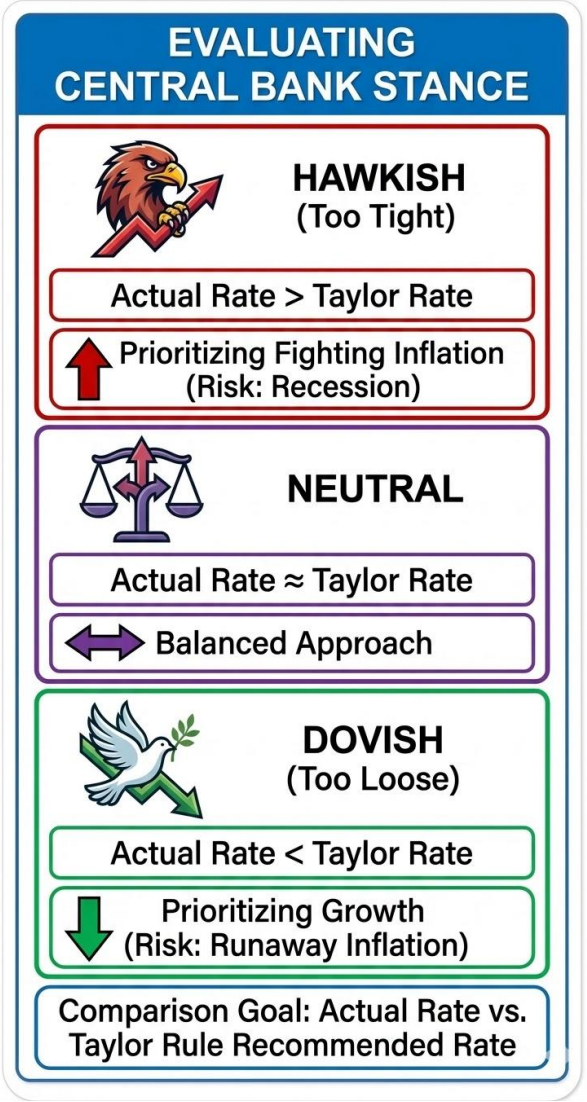
$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5(y_{gap})$$

- Let's translate the math into English:
- **The Base:** Start with the neutral rate (2%) plus current inflation.
- **The Inflation Penalty:** If current inflation is *higher* than the 2% target, add a penalty (raise rates to cool it down).
- **The Growth Penalty:** If the economy is growing *faster* than its potential (positive GDP gap), add a penalty (raise rates to stop overheating). If it's in a recession (negative gap), subtract (lower rates).



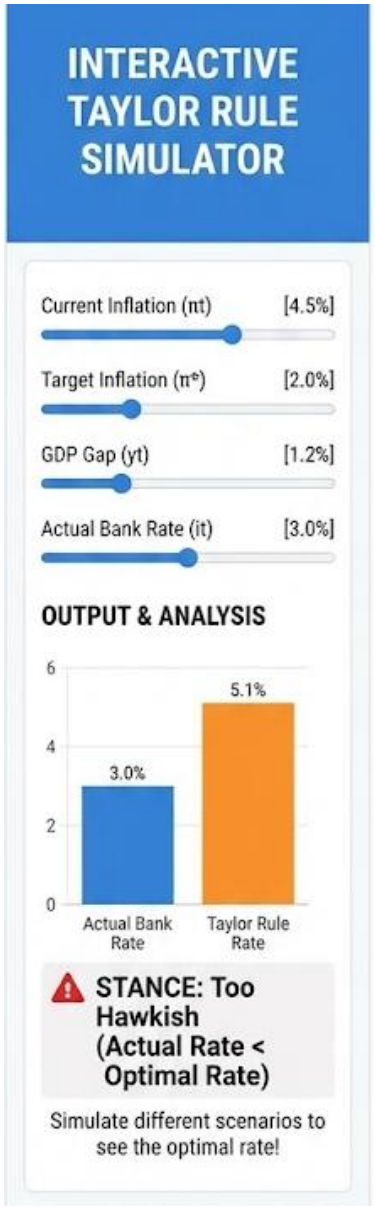
# Evaluating Central Bank Stance

- We can compare the *Actual* rate set by the bank to the *Taylor Rule Recommended* rate.
- **Hawkish (Too Tight):** Actual Rate > Taylor Rate. The bank is prioritizing fighting inflation, risking a severe recession.
- **Dovish (Too Loose):** Actual Rate < Taylor Rate. The bank is prioritizing growth, risking runaway inflation.
- **Neutral:** Actual Rate ≈ Taylor Rate.



# Interactive Taylor Rule Simulator

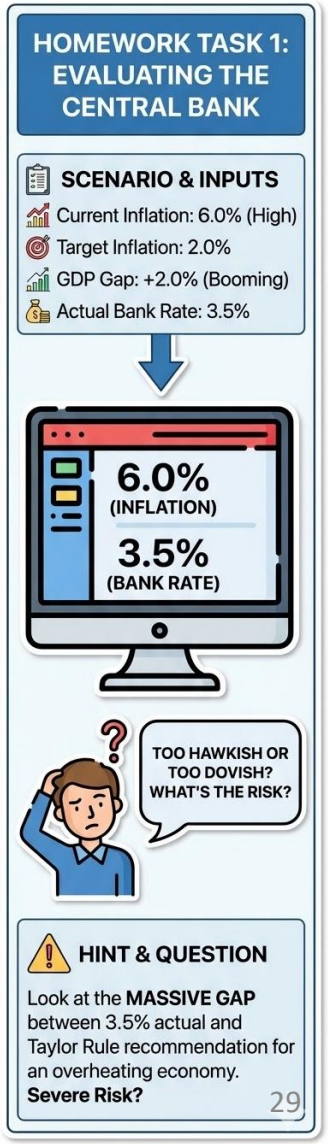
- In the notebook, you will use ipywidgets to play Central Bank Governor.
- **Sliders:** You input Current Inflation, Target Inflation, GDP Gap, and the Actual Bank Rate.
- **Output:** The Python script calculates the optimal rate and tells you if your current stance is "Too Hawkish," "Too Dovish," or "Just Right."



Part 7: Homework & Conclusion

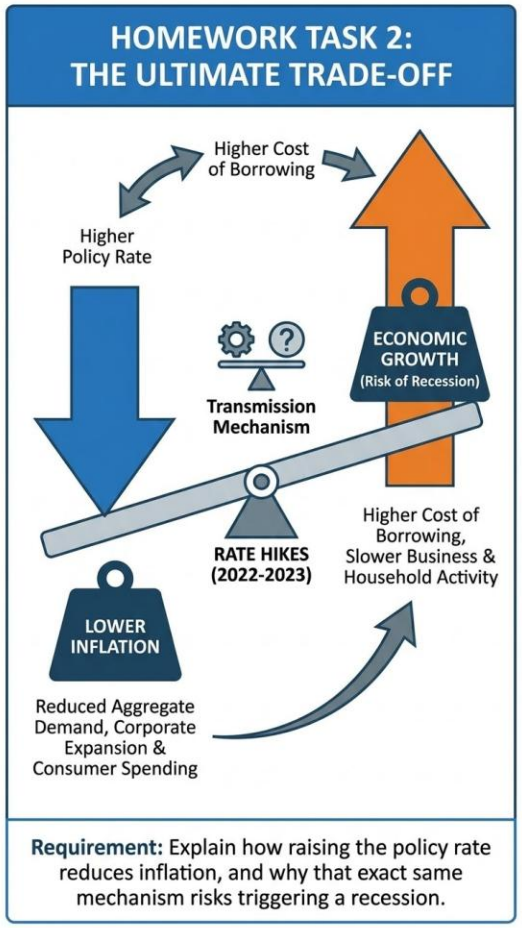
# Homework Task 1: Evaluating the Central Bank

- **Scenario:**
  - Current Inflation: 6.0% (High)
  - Target Inflation: 2.0%
  - GDP Gap: +2.0% (Booming)
  - Actual Bank Rate: 3.5%
- **Task:** Run these inputs through the notebook simulator.
- **Question:** Is the central bank "Too Hawkish" or "Too Dovish"? What is the severe risk of their current stance?



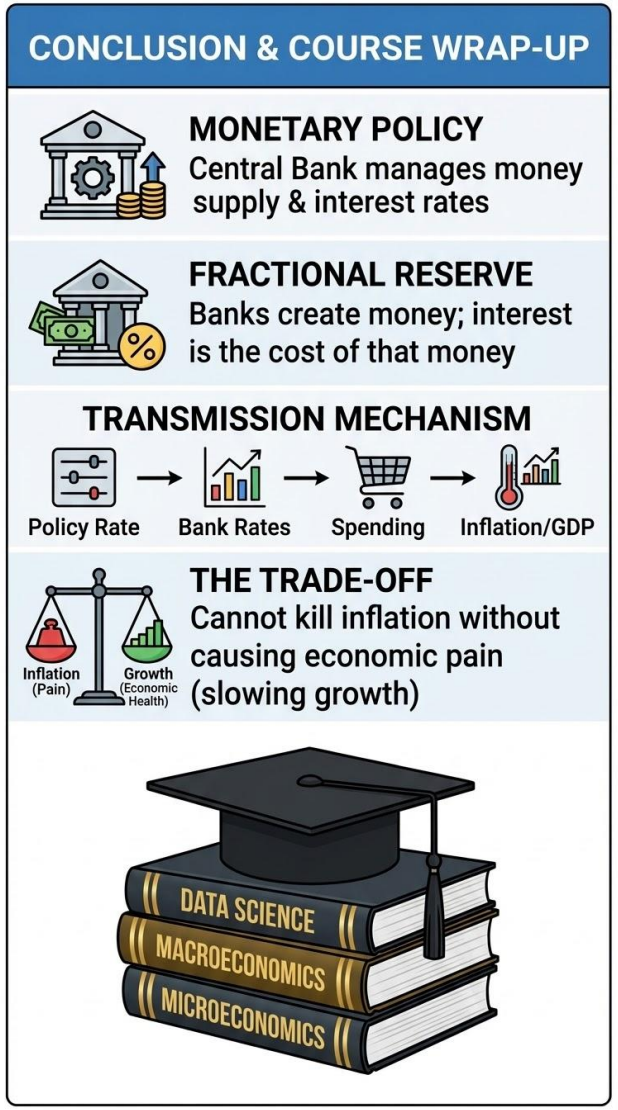
# Homework Task 2: The Ultimate Trade-off

- **Task:** Explain in your own words the **Transmission Mechanism** of the 2022-2023 rate hikes.
- **Requirement:** You must explain *how* raising the policy rate reduces inflation, and *why* that exact same mechanism risks triggering a recession.
- **Key Concepts to include:** Cost of borrowing, Aggregate Demand, Corporate Expansion, Consumer Spending.



# Conclusion & Course Wrap-up

- **Monetary Policy:** The Central Bank manages the money supply and interest rates.
- **Fractional Reserve:** Banks create money; interest is the cost of that money.
- **Transmission Mechanism:** Policy Rate → Bank Rates → Spending → Inflation/GDP.
- **The Trade-off:** You cannot kill inflation without causing economic pain (slowing growth).



# Let’s check out our Notebook

